

Stuff to know before beginning the chapter

⇒ standard conditions [for a fair comparison between enthalpy changes, we need to use the same conditions]

- 1) pressure of 101 kPa (approximately atm)
- 2) temperature of 293K (25°C)



symbol used to indicate that an enthalpy change occurred under standard conditions

⇒ enthalpy changes of different types of reactions

- 1) enthalpy change of reaction $[\Delta H_r]$ → ΔH of a reaction that occurs exactly like its balanced chemical equation
- 2) enthalpy change of formation $[\Delta H_f]$ → ΔH when 1 mol of compound is formed from its constituent elements
- 3) enthalpy change of combustion $[\Delta H_c]$ → ΔH when 1 mol of a substance burns completely in oxygen
- 4) enthalpy change of neutralization $[\Delta H_{\text{neut}}]$ → ΔH when 1 mol of water is produced from an acid+base reaction

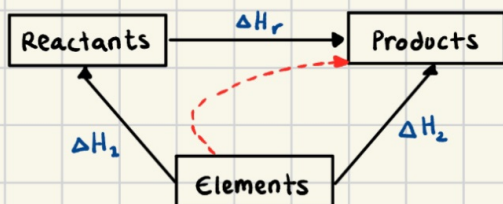
under standard conditions

Hess's Law

→ states that the total enthalpy change for a chemical reaction is the same, regardless of the route or number of steps taken, as long as initial & final conditions are the same



enthalpy change of formation :



$$\Delta H_2 = \Delta H_1 + \Delta H_r$$

direct route indirect route

$$\Delta H_r = \Delta H_2 - \Delta H_1$$

23 - Chemical energetics

09/12/25

ENTHALPY CHANGE OF ATOMISATION (ΔH_{at})

- enthalpy change when 1 mol of gaseous atoms is formed from its element in its standard state
- always endothermic



LATTICE ENERGY (ΔH_{latt})

- enthalpy change when 1 mol of an ionic compound is formed from its gaseous ions under standard conditions
- always exothermic



FIRST ELECTRON AFFINITY

- enthalpy change when 1 mol of electrons is added to 1 mol of gaseous atoms to make 1 mol of gaseous anions
- first EA is always exothermic & second EA onwards is always endothermic



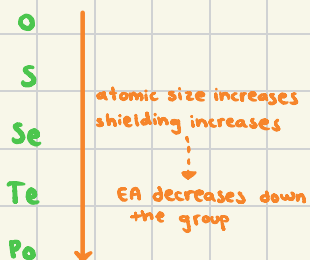
energy is required to overcome the repulsion b/w the anion & the electron

factors affecting electron affinities of elements:

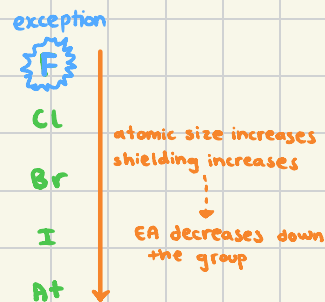
atomic size	↑	added e^- is further from nucleus	weaker attractive force b/w e^- & nucleus	lower EA
nuclear charge	↑	greater attractive force b/w e^- & nucleus		higher EA
electron shells	↑	greater shielding of added e^-		lower EA

trends in the EA of Group 6 & 7 elements:

GROUP 6:



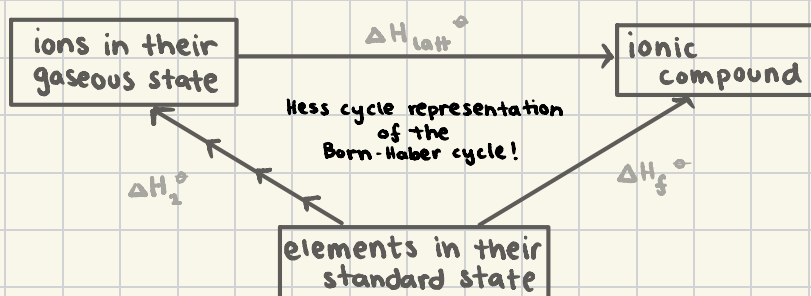
GROUP 7:



* EA of fluorine is less than that of chlorine & that is because fluorine has a very small atomic radius & high electron density which causes a greater electron repulsion

BORN - HABER CYCLE

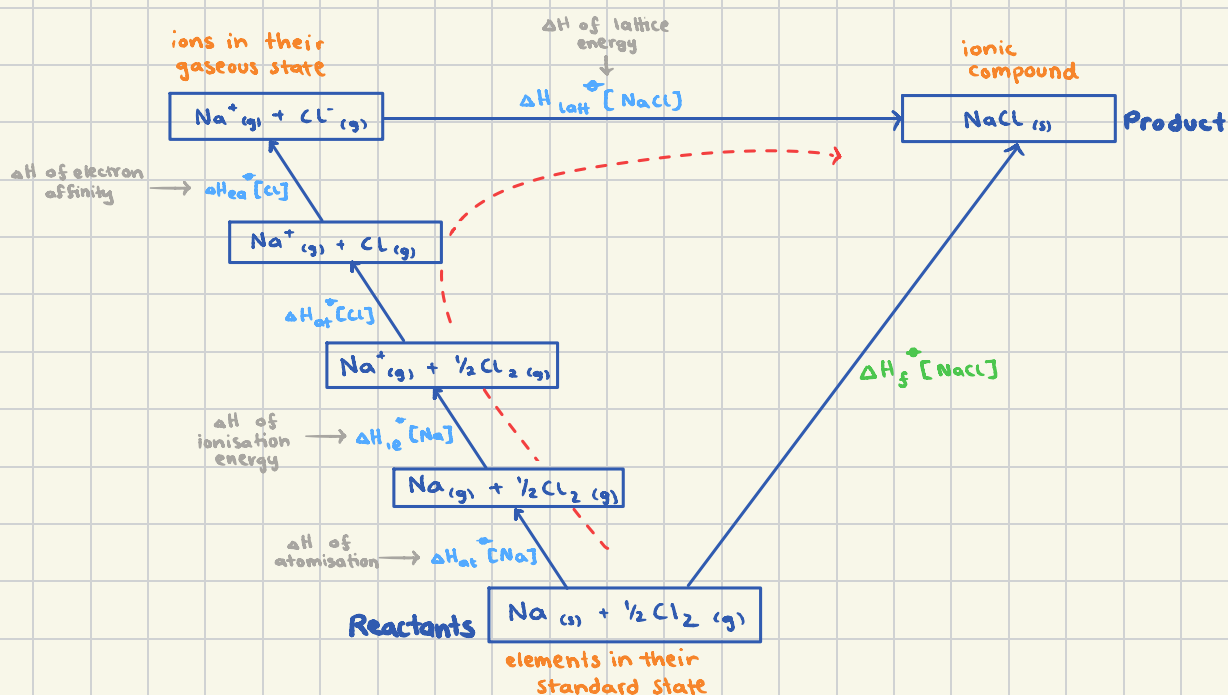
↳ an energy cycle used to calculate lattice energy



THIS ISN'T A DIRECT PROCESS THOUGH!

↳ to construct a Born-Haber cycle:

- ① atomise metal
- ② ionise metal — use ionisation energy
- ③ atomise non-metal
- ④ ionise non-metal — use electron affinities



$$\therefore \Delta H_f^\circ [\text{NaCl}] = \Delta H_{\text{at}} [\text{Na}] + \Delta H_{\text{ie}} [\text{Na}] + \Delta H_{\text{at}} [\text{Cl}] + \Delta H_{\text{ea}} [\text{Cl}] + \Delta H_{\text{latt}} [\text{NaCl}]$$

The effect of ionic charge & ionic radius on lattice energy:

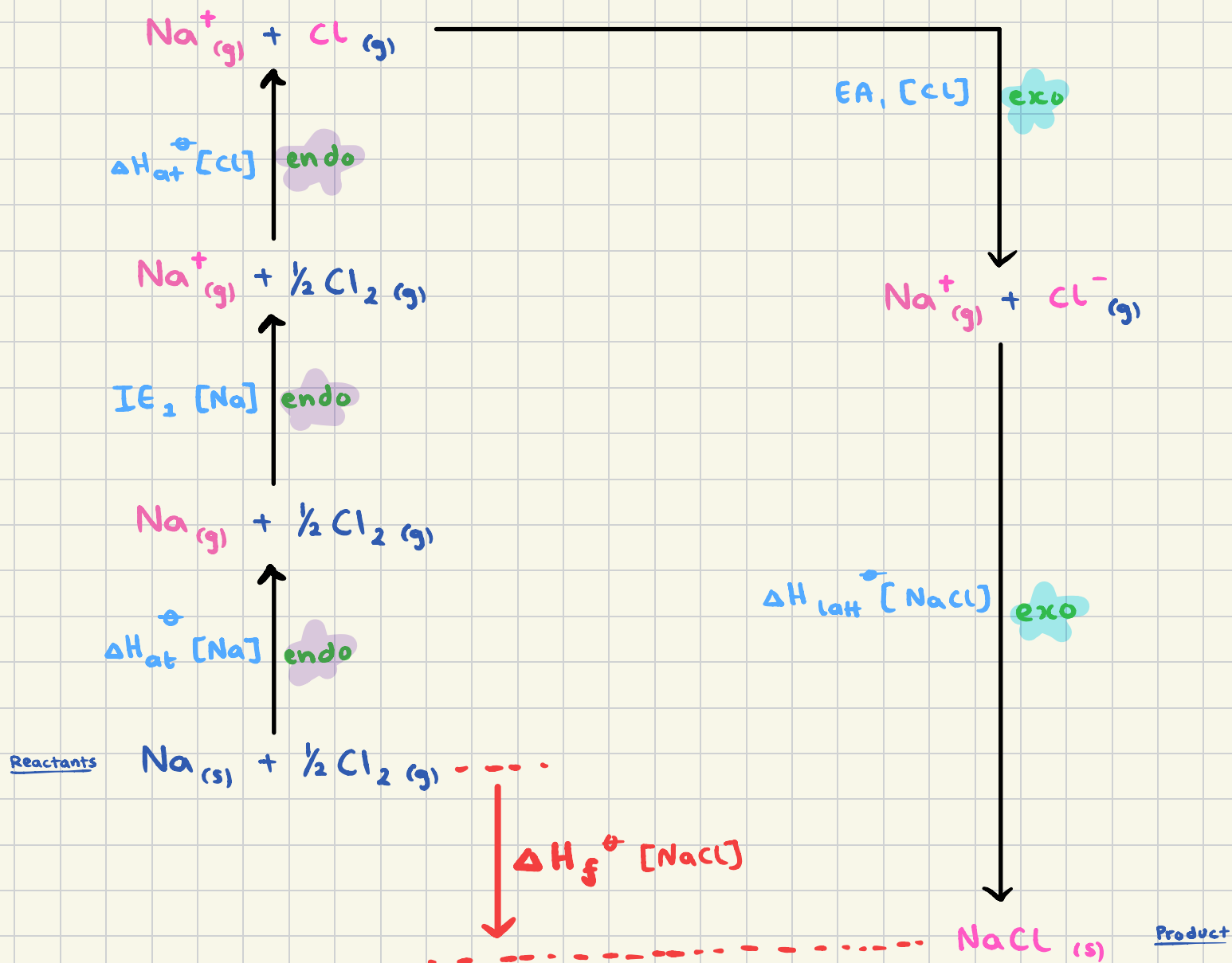
ionic charge ↑

- ↳ stronger electrostatic forces
- ↳ more energy released when lattice formed
- ★ more exothermic lattice energy
 - ↳ more negative
 - ↳ greater numerical magnitude

ionic radius ↑

- ↳ lower charge density
- ↳ weaker attraction b/w ions
- ↳ weaker electrostatic forces
- ★ less exothermic lattice energy
 - ↳ less negative
 - ↳ lower numerical magnitude

⇒ the reaction displayed by the Born-Haber cycle



$$\therefore \Delta H_f^\circ[\text{NaCl}] = \underbrace{\Delta H_{\text{at}}[\text{Na}] + \Delta H_{\text{ie}}[\text{Na}] + \Delta H_{\text{at}}[\text{Cl}] + \Delta H_{\text{ea}}[\text{Cl}]}_{\Delta H_1^\circ} + \Delta H_{\text{latt}}[\text{NaCl}]$$

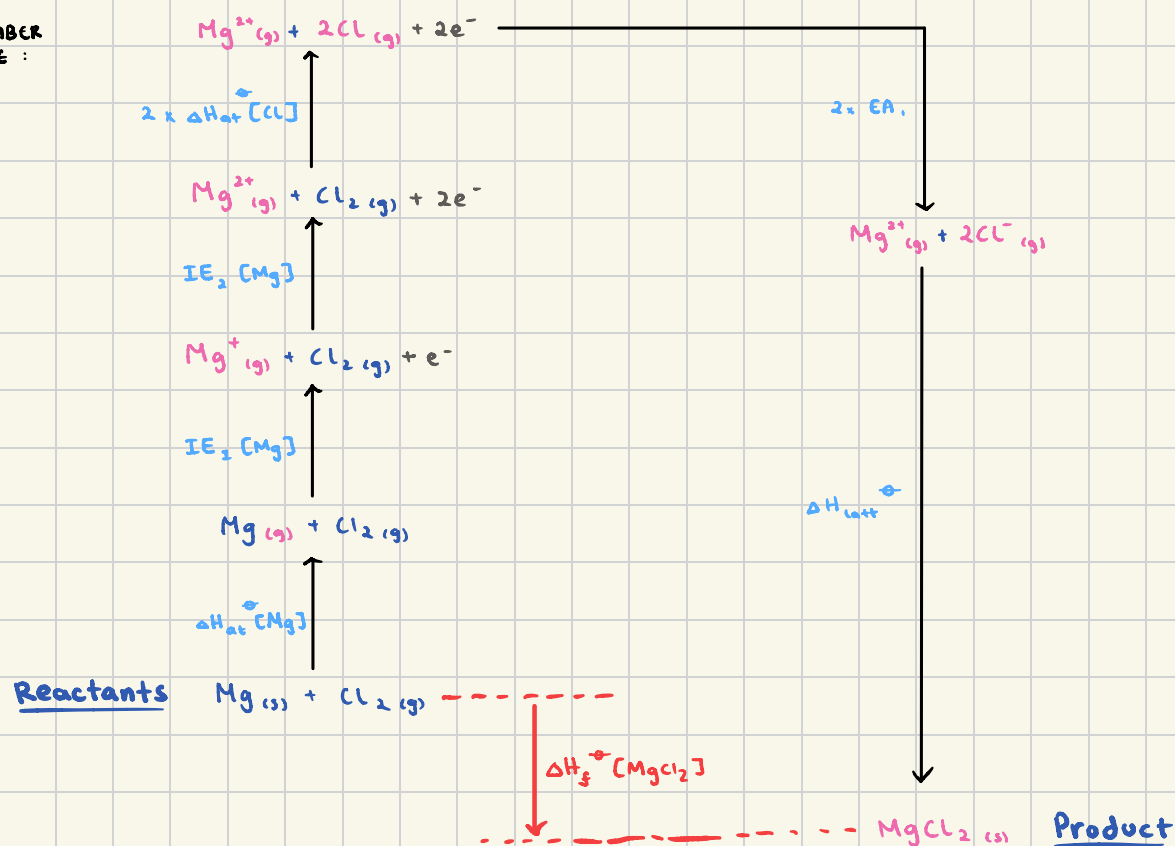
$$\therefore \Delta H_{\text{latt}} = \Delta H_f^\circ - \Delta H_1^\circ$$

e.g

OVERALL CON. :



BORN-HABER
CYCLE :



Calculate the lattice energy :

given that ...

$$\Delta H_{\text{at}}^{\circ}[\text{Mg}] = +148 \text{ kJ mol}^{-1}$$

$$\Delta H_{\text{at}}^{\circ}[\text{Cl}] = +122 \text{ kJ mol}^{-1}$$

$$\Delta H_{\text{f}}^{\circ}[\text{MgCl}_2] = -641 \text{ kJ mol}^{-1}$$

$$\text{IE}_1[\text{Mg}] = +736 \text{ kJ mol}^{-1}$$

$$\text{IE}_2[\text{Mg}] = +1450 \text{ kJ mol}^{-1}$$

$$\text{EA}[\text{Cl}] = -348 \text{ kJ mol}^{-1}$$

① Magnesium ion is Mg^{2+} so first & second IE need to be accounted for!



$$\text{IE}_1 = +736 \text{ kJ mol}^{-1}$$

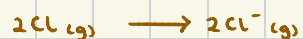


$$\text{IE}_2 = +1450 \text{ kJ mol}^{-1}$$

② Two chloride ions present so ΔH_{at} & EA_1 need to be multiplied by 2!



$$2\Delta H_{\text{at}} = 2 \times (+122) = +244 \text{ kJ mol}^{-1}$$



$$2\text{EA}_1 = 2 \times (-348) = -696 \text{ kJ mol}^{-1}$$

③ use Hess's law to calculate lattice energy

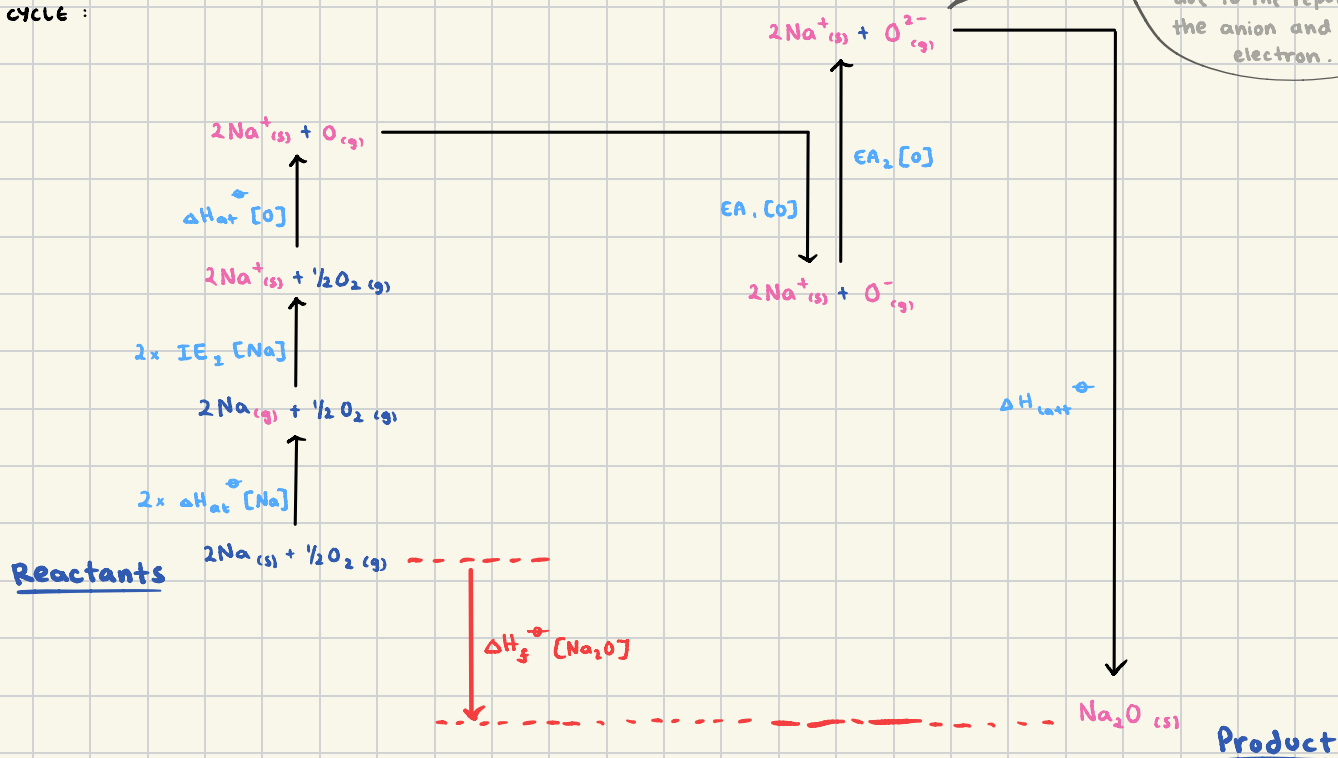
$$\Delta H_{\text{latt}} = \Delta H_{\text{f}} - \Delta H_1$$

$$\Delta H_{\text{latt}} = \Delta H_{\text{f}} - \{ \Delta H_{\text{at}}[\text{Mg}] + \text{IE}_1[\text{Mg}] + \text{IE}_2[\text{Mg}] + 2(\Delta H_{\text{at}}[\text{Cl}]) + 2(\text{EA}[\text{Cl}]) \}$$

$$\Delta H_{\text{latt}} = (-641) - \{ (148) + (736) + (1450) + (244) + (-696) \}$$

$$\Delta H_{\text{latt}} = -2523 \text{ kJ mol}^{-1}$$

e.g

BORN-HABER
CYCLE :

Calculate the lattice energy :

given that ...

* $\Delta H_{\text{at}}[\text{Na}] = +109 \text{ kJ mol}^{-1}$

* $\text{IE}_1[\text{Na}] = +494 \text{ kJ mol}^{-1}$

* $\Delta H_f[\text{Na}_2\text{O}] = -416 \text{ kJ mol}^{-1}$

* $\text{EA}_1[\text{O}] = -142 \text{ kJ mol}^{-1}$

* $\Delta H_{\text{at}}[\text{O}] = +248 \text{ kJ mol}^{-1}$

$\text{EA}_2[\text{O}] = +844 \text{ kJ mol}^{-1}$

① Two sodium ions present so ΔH_{at} & IE_1 need to be multiplied by 2!

$$2\Delta H_{\text{at}} = 2 \times (+109) = +218 \text{ kJ mol}^{-1}$$



$$2\text{IE}_1 = 2 \times (+494) = +988 \text{ kJ mol}^{-1}$$

② Oxygen ion is O^{2-} so first & second EA need to be accounted for!

$$\text{EA}_1 = -142 \text{ kJ mol}^{-1}$$



$$\text{EA}_2 = +844 \text{ kJ mol}^{-1}$$

③ use Hess's law to calculate lattice energy

$$\Delta H_{\text{latt}} = \Delta H_f - \Delta H_2$$

$$\Delta H_{\text{latt}} = \Delta H_f - \{ 2(\Delta H_{\text{at}}[\text{Na}]) + 2(\text{IE}_1[\text{Na}]) + \Delta H_{\text{at}}[\text{O}] + \text{EA}_1[\text{O}] + \text{EA}_2[\text{O}] \}$$

$$\Delta H_{\text{latt}} = (-416) - \{ (218) + (988) + (248) + (-142) + (844) \}$$

$$\Delta H_{\text{latt}} = -2572 \text{ kJ mol}^{-1}$$

ENTHALPY CHANGE OF HYDRATION (ΔH_{hyd})

- enthalpy change when 1 mol of gaseous ions dissolve in water to form a very dilute solution
- always exothermic



Effect of ionic charge & ionic radius on enthalpy change of hydration:

IONIC CHARGE $\uparrow$

L stronger electrostatic forces b/w ion & H_2O

★ more exothermic / -ve enthalpy change of hydration

IONIC RADIUS $\uparrow$

L weaker electrostatic forces b/w ion & H_2O

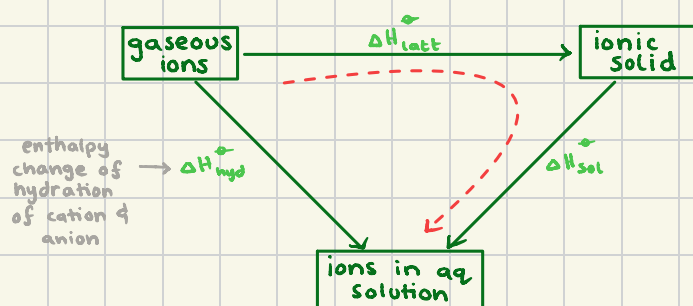
★ less exothermic / +ve enthalpy change of hydration

ENTHALPY CHANGE OF SOLUTION (ΔH_{sol})

- enthalpy change when 1 mol of an ionic solid dissolves in water to form a very dilute solution
- can be exothermic or endothermic



ENERGY CYCLE RELATING ΔH_{sol} , ΔH_{hyd} & ΔH_{latt}



Based on Hess's Law:

$$\Delta H_{\text{hyd}}(\text{M}^+) + \Delta H_{\text{hyd}}(\text{X}^-) = \Delta H_{\text{latt}} + \Delta H_{\text{sol}}$$

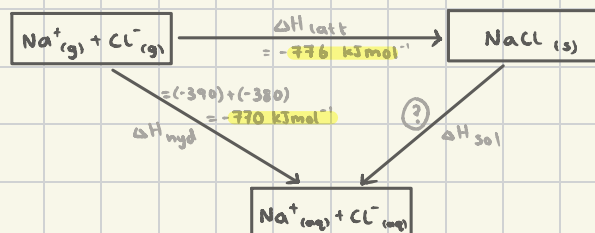
Q:

Find the enthalpy change of solution of NaCl

given that ... Lattice energy of NaCl = -776 kJ mol^{-1}

$$\Delta H_{\text{hyd}}(\text{Na}^+) = -390 \text{ kJ mol}^{-1}$$

$$\Delta H_{\text{hyd}}(\text{Cl}^-) = -380 \text{ kJ mol}^{-1}$$



$$\therefore \Delta H_{\text{hyd}} = \Delta H_{\text{latt}} + \Delta H_{\text{sol}}$$



$$\Delta H_{\text{sol}} = \Delta H_{\text{hyd}} - \Delta H_{\text{latt}}$$

$$\Delta H_{\text{sol}} = (-770) - (-776)$$

$$\Delta H_{\text{sol}} = +6 \text{ kJ mol}^{-1}$$

ENTROPY (S):

↳ the number of possible arrangements of particles & their energy in a given system

↳ it reflects the degree of disorder or randomness

* higher entropy means a greater number of possible arrangements & higher disorder

PREDICTING THE SIGN OF ENTROPY CHANGES (ΔS)

1) $\text{H}_2\text{O}_{(s)} \longrightarrow \text{H}_2\text{O}_{(l)}$ entropy change is $[+ve]$ because there are more ways of arranging energy in a liquid than a solid

2) Temperature $\uparrow$ entropy change is $[+ve]$ because there are more ways of arranging energy when particles have greater E_k

Temperature $\downarrow$ entropy change is $[-ve]$ because there are fewer ways of arranging energy when particles have lower E_k

3) $2\text{NO}_{(g)} + \text{O}_{2(g)} \longrightarrow 2\text{NO}_{2(g)}$ entropy change is $[-ve]$ because there are fewer ways of arranging energy in the products than in reactants because the number of moles of gas molecules decreases

CALCULATING ENTROPY CHANGE (ΔS)

$$\underbrace{\Delta S^\ominus}_{\text{Standard entropy change}} = \underbrace{\sum S^\ominus}_{\text{sum of}} (\text{products}) - \sum S^\ominus (\text{reactants})$$

① For the equation: $\text{N}_2(g) + 3\text{H}_2(g) \longrightarrow 2\text{NH}_3(g)$, calculate the entropy change

given that ... $S^\ominus[\text{N}_{2(g)}] = 191.5 \text{ J K}^{-1} \text{ mol}^{-1}$

$S^\ominus[\text{H}_{2(g)}] = 130.6 \text{ J K}^{-1} \text{ mol}^{-1}$

$S^\ominus[\text{NH}_{3(g)}] = 192.3 \text{ J K}^{-1} \text{ mol}^{-1}$

$$\therefore \Delta S^\ominus = [2(192.3)] - [(191.5) + 3(130.6)]$$

$$\Delta S^\ominus = 384.6 - [191.5 + 391.8]$$

$$\Delta S^\ominus = 384.6 - 583.3$$

$$\Delta S^\ominus = -198.7 \text{ J K}^{-1} \text{ mol}^{-1}$$

GIBBS FREE ENERGY CHANGE (ΔG):

↳ a quantity that determines the spontaneity of a process

↳ combines enthalpy (ΔH) & entropy (ΔS) changes

GIBBS EQUATION:

$$\Delta G^\ominus = \Delta H^\ominus - T \Delta S^\ominus$$

Labels under the equation:

- $\Delta G^\ominus$: standard Gibbs free energy change
- $\Delta H^\ominus$: standard enthalpy change
- T : Temp in K
- $\Delta S^\ominus$: standard entropy change

Q Calculate $\Delta G^\ominus$

Given that ...

$$\Delta H^\ominus = -92.2 \text{ kJ mol}^{-1}$$

$$\Delta S^\ominus = 120.5 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$T = 298 \text{ K}$$

* convert units: $\Delta S^\ominus = 120.5 \text{ J K}^{-1} \text{ mol}^{-1}$
 $= 0.1205 \text{ kJ K}^{-1} \text{ mol}^{-1}$

$$\Delta G^\ominus = \Delta H^\ominus - T \Delta S^\ominus$$

$$(?) = (-92.2) - (298)(0.1205)$$

$$\Delta G^\ominus = -92.2 - 35.9$$

$$\Delta G^\ominus = -128.1 \text{ kJ mol}^{-1}$$

FEASIBILITY OF A REACTION

- $\Delta G < 0$ the reaction is spontaneous (feasible) under the given conditions
- $\Delta G > 0$ the reaction is non-spontaneous (not feasible) under the given conditions
- $\Delta G = 0$ the reaction is at equilibrium

★ if ΔG is positive, the reaction is NOT feasible & if it is negative, the reaction is feasible

EFFECT OF TEMPERATURE ON FEASIBILITY

		ΔH	
		-ve	+ve
ΔS	-ve	reaction IS feasible AT LOWER TEMP	reaction is NOT feasible
	+ve	reaction IS feasible	reaction IS feasible AT HIGHER TEMP

1) value of T is always positive

$$2) \Delta G^\ominus < 0 \rightarrow \Delta H^\ominus - T \Delta S^\ominus < 0$$

Q calculate the min. temp at which the following reaction is feasible: $\text{CuCO}_3(\text{s}) \rightarrow \text{CuO}(\text{s}) + \text{CO}_2(\text{g})$

given that: $\Delta H = 86 \text{ kJ mol}^{-1}$ & $\Delta S = 170 \text{ J K}^{-1} \text{ mol}^{-1}$
convert unit: $\Delta S = 0.170 \text{ kJ K}^{-1} \text{ mol}^{-1}$

$$\Delta H - T \Delta S < 0$$

$$T > \frac{\Delta H}{\Delta S}$$

$$T > \frac{86}{0.170}$$

$$\rightarrow T > 505 \text{ K}$$